

## Random walk theory and the weak-form efficiency of the US art auction prices

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### Abstract

We perform variance ratio tests based on non-parametric methods to detect the size of the random walk component of the US art auction prices. The past 134 years of the US art prices exhibit large transitory component (72%) and based on this, the random walk hypothesis does not hold. However, possibly due to sparse data before 1935 or due to institutional changes of the art market after World War II, we detect structural breakpoints and find that the random walk hypothesis and the weak-form efficiency of the US art market cannot be rejected at least for the past 64 years.

*JEL classification:* G12; G14; C14

*Keywords:* Art prices; Random walk; Asset pricing; Spectral density; Variance ratio

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## 1. Introduction

We investigate the US art market from the efficiency point of view. We aim to understand the price behavior of art works, mainly paintings sold in the most prestigious US auction houses. We test the random walk hypothesis applying a specially constructed US art index using the Mei Moses Fine Art Index and the US subindex of the Artprice Global Index family, which all together cover a 134-year-long period. We apply variance ratio tests on the merged index to detect the size of the random walk component. We find that the price components, that is the random walk and the temporary components are time varying. We divide the 134-year-long period into three well defined overlapping subperiods: 1935-2008; 1945-2008; 1973-2008. 1935 is chosen as a possible breakpoint, since Mei and Moses (2002) argue that spurious negative correlation in the index estimated by the Case and Shiller (1987) method can be severe before 1935 because of sparse data. 1945 is an obvious choice because of institutional changes in financial markets after World War II. Finally, 1973 is chosen because price estimates of auction houses are introduced at that time, which can have serious impact on the price formation process because the estimates are upward biased (see, e.g., Mei Moses, 2005).

Applying the Cochrane (1988) variance ratio estimation, the random walk hypothesis can be rejected for the whole sample period. However, if we exclude the first 60 years from the data, the random walk hypothesis cannot be rejected. Different explanations can be given for the time-varying extent of price components: this can be due to the sparse data before 1935; the institutional changes in the post-World War II era.

Constructing the paper, as a first step in Section 2 we review some empirical works on efficient market hypothesis which use similar methodology to ours. We revisit the existing literature of the art market efficiency. In Section 3 we present the data used for our empirical

analyses; we show why it is necessary to construct a new art index. In Section 4 we introduce the methodology applied to measure the extent of the random walk component in the art prices. In Section 5 we demonstrate the time varying behavior of the components and in Section 6 we conclude the paper with some summary remarks.

## **2. Literature review**

The early tests of stock market efficiency surveyed in Fama (1970) found little evidence against the efficient market hypothesis. Poterba and Summers (1987) find positive autocorrelation over short horizons and negative autocorrelation over long horizons in international stock returns. Using variance ratio tests they show that there is a transitory component in stock prices which accounts for more than half of the variance of monthly stock returns. They sketch two possible reasons of the mean reverting property in stock prices: 1, time-varying expected returns; 2, slowly-decaying “price fads” possible because of noise trading. They argue that an approximately 50% transitory component in stock prices is difficult to explain by risk factors. Fama and French (1988) argue that the stochastic evolution of investment opportunities causes time-varying expected returns, generating negative autocorrelations in annual stock returns. Supposing that there are shocks to the expected returns which are independent of the rational forecasts of dividends, that is, the shock has no long-term effect on asset prices. Such a shock causes slowly decaying component in stock prices and long term negative autocorrelation in stock returns since the shock to the expected returns must be offset by the opposite adjustment of the market prices. The measured stationary component is more significant for small companies and for the period before 1940 (see Fama and French, 1988).

Keynes (1936) stresses that “all sorts of considerations enter into market valuation which are in no way relevant to the prospective yield”. If we accept this assertion and suppose that there is a limit of mispricing, it must induce mean reversion in asset prices through speculation forces. Merton (1987) also argues that the presence of negative autocorrelation indicates failure of market valuations. DeBondt and Thaler (1993) state that the mean reversion in stock prices is a result of investors’ overreaction.

There are numerous papers dealing with testing efficiency, but these studies mainly focus on the stock, debt, FX and commodity markets. The efficiency tests of the art market predominantly focus on equilibrium asset pricing, underperformance of masterpieces and the violation of the law of one price. Pesando (1993) using repeat sales regressions and semiannual data finds no evidence that print masterpieces (defined as the most expensive ten and twenty percent) outperform the market. Goetzmann (1996) documents no underperformance of masterpieces. Mei and Moses (2002) obtain contrary results that the masterpieces defined as expensive paintings significantly underperform the art market as a whole. A ten percent increase in art prices is expected to lower the future annual returns by 0.1 percent. They argue that the underperformance of the masterpieces is similar to the small firm effect documented by Banz (1981) and Moller and Zilca (2008). Although, it is possible that masterpieces are less risky and more liquid, they consider that the existing mean reversion in art returns and the underperformance of masterpieces is due to this property. Another reasoning is that the mean reversion is a result of overreaction of collectors in a similar way as described in DeBondt and Thaler (1993). Locatelli Biely and Zanola (2002) find that the portfolio constructed by expensive sculptures outperforms the inexpensive/middle portfolio in the period 1987-1995.

Pesando's (1993) results show mixed evidence on violation of the law of one price; during the period 1989-1992 prices for identical prints were higher in the USA than in London or Europe, but on the other hand, during 1977-1988 he does not find such evidence. There is evidence, though, that the works of art by American artists realize higher prices in the United States. He shows that identical print prices tend to be higher at certain auction houses. In a more recent paper of Pesando and Shum (2007) they find that violation of the law of one price that occurred between prices realized at Christie's and at Sotheby's in New York during the period 1977-1992 did not persist, the results reversed during 1993-2004. Pesando and Shum show that the only statistically significant difference in realized prices identified in the period 1977-1992 that persisted to the later period 1993-2004 is the higher prices realized in the United States than in Europe. Mei and Moses (2002) find significant departure from the law of one price as the works of art bought at other than Christie's and Sotheby's US auctions yield significantly higher return and the prices of Sotheby's tend to be higher than those at Christie's on average. Locatelli Biey and Zanola (2002) show that sculpture prices recorded at Sotheby's are higher than those at Christie's, and prices tend to be higher in New York than in London for the period 1987-1995. Beggs and Graddy (2005) also show that art prices tend to be higher in New York than in London for the period 1980-1990; however, they argue it is due to quality differences between works of art auctioned in the two cities.

Applying the Capital Asset Pricing Model (CAPM) Pesando (1993) finds no evidence that art market significantly out or underperforms the S&P 500. He finds positive autocorrelation in the short and negative in the long run returns consistent with the findings considering the stock market. Mei and Moses (2002) apply one-factor CAPM on art returns and find similar evidence

as Goetzmann (1993) that the stock market drives the art market since the demand for art increases with the wealth of collectors.

A more recent study by Mei and Moses (2005) investigates the impact of price estimates on art price formation from a behavioral point of view. They show that the price estimates of the auction houses are biased upward and it has impact on investors because they are credulous consistent with studies on stock markets. Higher high estimate results in higher sales price and generates lower future returns. They argue that the upward bias is persistent over a period of 30 years because of agency problems and rational learning cannot eliminate it. Similarly on the capital markets, Womack (1996) finds that security analysts are biased, stock recommendations favor buys over sells. Capstaff et al. (1998) find that earnings forecasts are generally optimistic especially in the long run. Michaely and Womack (1999) show that information is manipulated due to agency problems between the investment banking industry and the investors. Teoh et al.(1998) argue that manipulation has impact on future prices: the long-run underperformance of IPOs is positively related to the manipulation of information. Mei and Moses (2005) find that collectors intend to pay more for masterpieces because they have less idiosyncratic risk. It is not consistent with rational pricing since unsystematic risk is diversifiable.

There is some evidence of ‘irrational exuberance’ in art prices as documented by Pesando and Shum (2007). The sales of the Picasso print collection by Sally and Victor Ganz auctioned at Christie’s New York in 1997 provides an illuminating example of how enthusiastic bidders may push art prices to levels well beyond their fundamental values, as evidenced by the sharply lower prices realized by the five Picasso prints in their subsequent appearances at auction. Goetzmann and Spiegel (1995) argue that collectors are disposed to overpaying for paintings because of their private value of art works and they are able to forego of some of the monetary returns in

exchange for viewing pleasure. On the one hand, this result suggests that not all of the buyers are speculators in the art market. However, the run-up in prices caused by aggressive bidding of collectors might be corrected in the subsequent auction causing mean-reversion in art prices as it was the case in the Ganz collection.

There are several studies dealing with the analysis of the randomness of art prices. Frey and Pommerehne (1989) find that real art returns do not follow a normal distribution and conclude that prices seem not to follow a pure random process in their data for the period 1635-1949 contrary to Baumol's (1986) findings for the period 1652-1961.

### **3. Data**

We use Mei Moses Fine Art Index from 1875 to 1991 and from 1992 to 2008 we extend the data with the Artprice USA Index, hereinafter the resulted index series is the Art Price Index (API).<sup>1</sup> Poterba and Summer (1987) show that the random walk tests induce low power whatever test is under consideration if short time series is available and the only opportunity to raise the power of the test is to use as many data as it is possible. For this reason we combine the Mei Moses Fine Art Index with the Artprice US index. It is essential to obtain as reliable a test as possible, even if the eligible art pieces included in the two indexes and their methodology are obviously not identical; however, in the literature it is not unusual to combine two different indexes when both track the same market and use similar methodology. Both indexes are constructed by repeat sales regression (RSR) method, first introduced by Bailey et al. (1963) for real estate price indexes, but widely used for the art market as well (see Baumol, 1986; Pesando,

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<sup>1</sup> The current value file of the Mei Moses index is not available; however, we can use the index values published at the end of the paper of Mei Moses (2002). The Mei Moses Fine Art Index is available until 1999; however, the Artprice USA index contains more auction results thus it is more precise, so from 1992 (this is the time from it is available) we use this as the art market proxy.

1993; Goetzmann, 1993, 1996; Pesando and Shum, 1996; Mei and Moses, 2002). The Mei Moses Fine Art Index includes all the art price records of American, Nineteenth-century, Old Master, Impressionist and Modern paintings sold at Sotheby's and Christie's (and their predecessor firms) from 1950 to 1999. The index is based on 4,896 price pairs for the period 1875-1999 (see Mei and Moses, 2002). If a painting has a provenance which indicates a publicly available sales pre 1950, they collect the sales price wherever the artwork was sold.

Artprice.com is currently using sales data from more than 2,900 auction houses worldwide and tracking the art market as a whole. They provide data in categories of fine art, design, antique furniture, works of art, collectibles, jewelry, clocks, ceramics, glass, etc. The Artprice Global Index (AGI) family includes paintings, prints, sculptures, photographs and drawings. At first sight it can be striking to extend the Mei Moses index with the AGI US index since it includes not just paintings but other works of art as well. The AGI has a subindex for paintings and the correlation between the AGI and the AGI Painting subindex is 99% for the period from July 1990 to December 2008. The other subindexes also have very high correlation with the art market, above 92% except photographs which have a correlation of 77% with the AGI. Based on the studied correlations, paintings drive the art market, which is not surprising since paintings have the highest market share of the art market. According to the Artprice methodology a repeat sales takes place if two works of art are sold sequentially which are by the same artist and their size, technique, materials, medium and the date of creation are matched.

### *3.1. Possible biases in the Art Price Index*

Our data are not free of selection and survivorship bias. The selection bias is rather an issue before only 1991 in our constructed index because the Mei Moses Art Index is based solely on Sotheby's and Christie's sales data which represent only a narrow subset of the art market transactions. However, in the AGI these biases are less severe because the eligible auction houses included in the index are in a much broader range, from small local auction houses to the most prestigious ones.

Survivorship bias also can be an issue in the Mei Moses Art Index as only the most successful items are included in the index. Solely the Sotheby's and the Christie's records are included in the Mei Moses Art Index; these auction houses handle only the highest quality works of art and if a painting falls out of fashion and it fails on an auction, it is very likely that it will not be included in the index again (see Goetzmann, 1993; Beggs and Graddy, 2005). Because of the selection and survivorship biases, auction transactions may not adequately reflect one of the most important elements of risk for the art investor, the stylistic risk. The prices of art pieces depend on the demand and the wealth of investors when they are up for sale. Since the repeat sales data contain only auction transactions, they necessarily focus upon art works that are in broad enough demand. Thus, the repeat sales data might not capture the price fluctuations of non high-end paintings or paintings which fall out of fashion (see Goetzmann, 1993). We have to note that if art is treated as an investment and if we exclude the works of unknown artists or young contemporary artists, whose paintings are generally low priced we can ensure that the consumption motivation of purchasers have no effect on our results (see Hodgson and Vorkink, 2004). For this motivation the Mei Moses Art Index is proper. On the other hand, we cannot

exclude buy-ins from the data if art pieces are treated as collaterals (see McAndrew and Thompson, 2007).

Our data might not be capable to track the price formation of the whole market because of the selection bias. There is evidence in the literature that price behaviors are different on the dealers' and on the auction market. There are institutional differences, auction market is centralized, transactions come in sequence following the sale order of the auction catalogue; the price is flexible and clears instantaneously the market. In the dealers' market transactions are decentralized and prices are determined by the seller (the gallery or the dealer) and they are sticky; however, the final prices are the result of a bargaining process. Despite the structural differences, price formations might not alter significantly since auction prices represent guideposts for collectors and professional art dealers (Frey and Pommerehne, 1989).

The RSR method has some advantages over another widely used method, the hedonic price index. The RSR is based upon price relatives of the same painting that controls the differing quality of the assets. Thus, it does not suffer from arbitrary specifications of a hedonic model. The drawback is that the index is constructed from multiple sales, which is only a subset of the available transactions. Goetzmann (1993) points out that the auction data truncates both the high end and the low end of the return distribution. Paintings that fall drastically after the purchase or are not in demand are generally not resold in auction; in addition, paintings that are donated to museums generally do not reappear. Furthermore, an owner's decision to sell a painting at auction may be determined by whether or not the painting has increased in value, Kliger and Kudryavtsev (2008) find similar results on the US stock market.

### 3.2. Art index methodology

The RSR can be estimated in the form

$$r_i = \ln\left(\frac{P_{i,s}}{P_{i,b}}\right) = \sum_{t=b_i+1}^{s_i} r_{i,t} = \sum_{t=b_i+1}^{s_i} \mu_t + \sum_{t=b_i+1}^{s_i} \varepsilon_{i,t}, \quad (1)$$

where  $r_i$  is the logreturn during the holding period for art piece  $i$ , that is the return accumulated between  $b_i$  (the time of the purchase) and  $s_i$  (the time of the sale),  $P_{i,b}$ ,  $P_{i,s}$  are the purchase and the sales price of the art work  $i$  respectively,  $\mu_t$  is the average return of the market in period  $t$  and  $\varepsilon_{i,t}$  is the disturbance term of the estimated regression (see Goetzmann, 1992). Mei and Moses (2002) estimate Equation (1) using the Weighted Repeat Sales (WRS) method by Case and Shiller (1987) because it adjusts for a downward bias in annual returns estimation due the logprice transformation (see Goetzmann, 1992).<sup>2</sup> The WRS estimator is a three-stage regression procedure; the first stage is identical to the OLS estimation of Equation (1); in the second stage the squared residuals are regressed on a constant and the time interval between the purchase and the sales; in the third stage the generalized least squares (GLS) regression is run by first dividing each observation by the square root of the forecast value of the second stage regression.<sup>3</sup> In this procedure the weights of the observations are inversely related to the holding periods of the art works.<sup>4</sup>

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<sup>2</sup> They also estimate the art index using GLS and the two-stage Bayesian estimation proposed by Goetzmann (1992). The correlation between the Case and Shiller (1987) procedure and the other two procedures is 0.970 and 0.917, suggesting that the results are quite robust. They also discover that the two-stage Bayesian estimates tend to have smaller estimation errors though they may be biased.

<sup>3</sup> The first-stage regression is the estimation proposed by Bailey et al. (1963).

<sup>4</sup> Goetzmann (1992) and Goetzmann and Peng (2001) thoroughly study the statistical properties of different RSR estimations. Goetzmann (1992) compares the OLS, GLS (the weights are the square root of the inverse of the holding period), WRS and different Bayesian methods and argues that RSR estimators underestimate the mean and overestimate the standard deviation of equally weighted index on monthly stock returns. He proposes a Bayesian approach for RSR estimation for daily stock returns; however, it generates spurious positive correlation of US monthly stock returns in his data. In the Goetzmann study the results are robust through different estimators, they are almost identical using monthly returns; however, the WRS procedure underperforms the others according to some statistical diagnostics. We can reckon on that the difference through the estimators are even less for the Mei Moses Fine Art Index since they use annual returns and the goodness of fitting increases with return horizon

Collins et al. (2009) argue that art indexes (either based on RSR or hedonic regressions) can be biased due to non-randomness in data selection. There is a high portion of unsold items on auctions which is excluded from the dataset. They propose the Heckman style correction for solving the sample selection bias in estimated regressions. Mei and Moses (2002) do not use this correction in estimating the art index so our data are not corrected against selection bias.<sup>5</sup>

Another approach is proposed by Locatelli Biey and Zanola (2005) who introduce a novel approach for estimating art indexes using the procedure of Carter Hill et al. (1997). They apply a hybrid model combining the RSR and the hedonic regression using the seemingly unrelated regression procedure. They show that the confidence interval for the estimated index is tighter than for either the RSR or the hedonic regression.

Estimating the net art returns seem to be problematic because transaction costs are excluded and in contrast to stocks and bonds, these can be quite high (as much as 25 percent of the value of the object, considering both buyer's and seller's premiums) and there is a significant theft and fire risk (and hence insurance costs) and cleaning costs involved in investing in art (see Ashenfelter and Graddy, 2003). On the other hand, works of art have some non-monetary dividend yield considering the aesthetic pleasure and the social status of the owner (Goetzmann, 1993).

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quadratically (see Webb 1981, Goetzmann 1992). On the other hand, the Case-Shiller (1987) procedure does not work well for stock indexes, but it does not mean that it is not adequate for infrequently traded goods like art works since this estimator is developed specially for such assets.

<sup>5</sup> We contacted Mei and Moses and the Artprice.com to obtain raw data on auction records to estimate a Heckman corrected RSR art index but unfortunately our requests have been rejected. We have been searching for other databases with similar coverage; however, we have found ones only with different geographical focus, for different collectible categories and for a much shorter period. The last is the main problem, because the longest possible time series is required if one aims to apply any kind of spectral density estimator to detect the size of the random walk component in stochastic processes.

### 3.3. Unit root tests of the API

We investigate whether or not the constructed art index follows random walk. The standard way of testing random walk is the Augmented Dickey-Fuller (ADF) test. The ADF test is not able to reject the unit root hypothesis at any usual significance level no matter we include a constant and a trend or just a constant.

[ Please insert Figure 1 here ]

Figure 1 shows that the US art market has an exponential upward trend during the past 130 years, especially from the middle of the 20<sup>th</sup> century so it is not striking that the unit root hypothesis cannot be ruled out based on the ADF test. We get a better view of the art prices, if we calculate real price index which is free of possible random walk component in the consumer price index (CPI). The CPI for the period 1875-2008 is obtained from Global Financial Data.

First, we take the logarithm of the art price index and then deduct the logarithm of the US CPI. We rerun the ADF tests on the log real index (Figure 2), and we find the null hypothesis still cannot be rejected at any usual significance level if only a constant term is included. It means the possible random walk component in the nominal index is not or only partly due to the possible random walk in the CPI.

[ Please insert Figure 2 here ]

Applying the ADF test including constant and linear trend, we find that the unit root nullity can be rejected at 95% level. This result suggests that the real art index is trend stationary. Summers (1986) show that these tests have low power against alternatives to the null of market efficiency (random walk) with constant required returns.

Poterba and Summers (1987) investigate the power of different random walk tests and find that variance ratios are among the most powerful tests and it has even more power than the Fama

and French (1988) regression based procedure. They argue that even the variance ratio test has very little power against particular alternatives and they show that the standard significance level choice of five percent is not appropriate. They propose to choose 40% significance level for one who wants to minimize the sum of Type I and Type II errors. Their results are based on Monte Carlo experiment and on the assumption of ARMA(1,1) process. They also show that variance ratio tests are among the most powerful ones detecting mean reversion.

Summers (1986), Poterba and Summers (1987), Cochrane (1988), Fama and French (1988), Lo and MacKinlay (1988), among others argue that unit root tests have very low power against stationary alternatives and it is difficult to reject a false null hypothesis of random walk. In the coming sections we will investigate art prices from the stationary, trend stationary and random walk point of view.

#### 4. Methodology

Let assume a pure random walk process with drift  $\alpha$  of the form

$$P_t = \alpha + P_{t-1} + \varepsilon_t, \quad (2)$$

where  $\varepsilon_t$  is a white noise process with constant variance and zero expected value. The variance of  $k$ -difference of this process is proportional to  $k$ :

$$\text{Var}(P_t - P_{t-k}) = k\sigma^2(\varepsilon). \quad (3)$$

Since  $P_t$  is a logprice index,  $P_t - P_{t-k}$  is the  $k$  holding period return. Then, if we assume a trend stationary process in the form of

$$P_t = at + \sum_{k=0}^{\infty} a_k \varepsilon_{t-k}, \quad (4)$$

where  $a_0 \equiv 0$ , the variances of  $k$ -difference should be constant at

$$\text{Var}(P_t - P_{t-k}) = 2\sigma^2(\varepsilon). \quad (5)$$

Stationary or trend stationary time series have lag-independent variances of  $k$ -difference. Using the above sketched approach, the process can be described better. We follow Cochrane (1988) procedure and plot  $(1/k)\text{Var}(P_t - P_{t-k})$  against lag  $k$  on Figure 3. If the art prices follow a pure random walk process the plot should be flat at the variance of first difference. We can see that is not the case but variances of  $k$ -difference do not approach zero, since after lag 11 the graph is flatter especially if we look at Figure 5. The empirical investigation of variances of  $k$ -difference shows that art prices follow neither a pure random walk, nor a pure stationary process. Based on Figures 3-7 the most likely is that art prices have two components: a stationary component which has effect up to approximately  $k=13$ , and a random walk component which is a permanent component so it does not die out.

[ Please insert Figure 3 here ]

[ Please insert Figure 4 here ]

#### *4.1. Separation of the random walk and the stationary component*

To estimate the extent of the two components, that is the random walk and the stationary components, we use a similar approach described in Cochrane (1988).

The theoretical background is based on the Beveridge-Nelson (1981) decomposition, that is, every first difference stationary process can be decomposed to a pure random walk and to a stationary process. Let  $P_t$  denote the log art index value at period  $t$  which is a first-difference

stationary process<sup>6</sup>. In this case  $\Delta P_t$ , the logreturns series has a moving average representation in the form

$$\Delta P_t = \alpha + \sum_{k=0}^{\infty} a_k \varepsilon_{t-k}, \quad (6)$$

where  $\alpha$  is a constant and  $\varepsilon_t$  is an i.i.d. standard normal innovation. The Beveridge-Nelson decomposition of  $P_t$  is as follows

$$P_t = Y_t + S_t, \quad (7)$$

where

$$Y_t = \alpha + Y_{t-1} + \left( \sum_{k=0}^{\infty} a_k \right) \varepsilon_t, \quad (8)$$

$$-S_t = \left( \sum_{k=1}^{\infty} a_k \right) \varepsilon_t + \left( \sum_{k=2}^{\infty} a_k \right) \varepsilon_{t-1} + \left( \sum_{k=3}^{\infty} a_k \right) \varepsilon_{t-2} + \dots \quad (9)$$

This decomposition is constructed in such a way that the expected value of  $P_{t+k}$  should be equal to the expected value of  $Y_{t+k}$ ; i.e.,  $Y_t$  is the permanent component and  $S_t$  has no impact on the long term expected value of  $P_t$ .

Let denote

$$\sigma_k^2 = k^{-1} \text{Var}(P_t, P_{t-k}), \quad (10)$$

that is the variance of  $k$ -difference divided by  $k$ . If  $P_t$  is a pure random walk,  $\sigma_k^2$  should be flat when the  $k$  lag is increasing. If  $P_t$  is a stationary process without random walk component  $\sigma_k^2$  should decline to zero as  $k$  approaches infinity.

Cochrane (1988) proves that the variance of  $k$ -difference can be estimated by Bartlett kernel which has known large sample properties:

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<sup>6</sup> We test the first-difference of  $P_t$  employing the ADF test and the null hypothesis can be rejected at any usual significance level.

$$\sigma_k^2 = \left( 1 + 2 \sum_{j=1}^{k-1} \frac{k-j}{k} \rho_j \right) \sigma_{\Delta P}^2, \quad (11)$$

where  $\sigma_{\Delta P}^2 = \text{Var}(P_t - P_{t-1})$ , that is the variance of one-period return and  $\rho_j = \text{Cov}(\Delta P_t - \Delta P_{t-j}) / \sigma_{\Delta P}^2$ , that is  $j$ -order autocorrelation of the one-period return. As  $k$  approaches infinity, that is enlarging the return horizon, the variances of  $k$ -year return approach the variance of the permanent component of  $P_t$  at large  $k$ , so that

$$\lim_{k \rightarrow \infty} \sigma_k^2 = \left( 1 + 2 \sum_{j=1}^{\infty} \rho_j \right) \sigma_{\Delta P}^2 = \sigma_{\Delta Y}^2. \quad (12)$$

Equation (12) estimates the spectral density at frequency zero (see e.g., Cochrane, 1988), which is the long run variance of the art index. In the very long run only the random walk component has effect on prices, the transitory component vanishes. As lag  $k$  approaches infinity, the variance of the art returns is constant at the variance of the permanent component.

Let denote  $VR_k$  the variance ratio statistic at lag  $k$  which is equal to

$$VR_k = \frac{\sigma_k^2}{\sigma_1^2}, \quad (13)$$

which shows the size of the random walk component in the price index (see Cochrane, 1988).

Figure 3 illustrates the Bartlett kernel estimation of the variance of  $k$ -differences of the investigated art price index. The variance of  $k$ -year return series become flatter from the lag 11 as  $k$  grows. Figure 4 shows the variance ratios of the art index; the random walk component is only 20.71% at lag 30. The variance ratio is more than two standard errors from unity so we can reject a null hypothesis of pure random walk at any usual significance level. Although, the variance ratio test is decisive on random walk hypothesis, a pure stationary process cannot be rejected since the variance ratio is less than two standard errors from zero.

We have reasons to suppose that a simple Bartlett estimation is biased in small samples. First of all, our 134 data points cannot be thought of as a large sample. Cochrane (1988) points out the Bartlett estimation is biased in small samples as  $1/k$  times the variance of  $k$ -period return declines toward zero as  $k$  approaches the sample size ( $T$ ). Cochrane proposes two types of correction; first, the drift term should be thought of as the sample mean of the first difference instead of re-estimating the mean at all  $k$ . Second, he uses degrees of freedom correction  $T/(T-k+1)$ . Applying the two types of correction, the estimation is the form of

$$\sigma_k^2 = \frac{T}{k(T-k)(T-k+1)} \sum_{j=k}^T \left[ P - P_{j-k} - \frac{k}{T}(P_T - P_0) \right]^2, \quad (14)$$

which is unbiased even in small samples (see Cochrane, 1988 for the proof).

[ Please insert Figure 5 here ]

[ Please insert Figure 6 here ]

The unbiased variances of  $k$ -period returns on Figure 5 and the variance ratios on Figure 6 are flatter than in the case of the Bartlett kernel estimation. The decline in  $VR_k$  after  $k=12$  in the Bartlett estimation can be observed possibly because of the above mentioned downward bias. In the case of the Cochrane estimation from lag 12, the curve is flat at the extent of the random walk component. Using the unbiased estimator the random walk component is almost 50% higher than in the case of the Bartlett estimation although the two estimations are not significantly different from each other<sup>7</sup>. The variance ratio at lag 30 using the Cochrane estimation is 29.43%.<sup>8</sup>

[ Please insert Figure 7 here ]

<sup>7</sup> The Bartlett estimation is within the one standard error band of the Cochrane estimation.

<sup>8</sup> The Cochrane estimation in small samples is upward biased at large  $k$ , we run Monte Carlo experiments to see this property. They are available upon request. In our sample  $k=30$  is a fairly good choice for maximum lag. See Table 1 where we demonstrate that the theoretical variance ratio in infinity is close to the value at lag 30.

Based on the Cochrane estimation, the variance ratios are more than two standard errors from unity and zero, so we can reject either a null hypothesis of pure random walk, or a stationary process without random walk component. The unbiased estimator declines up to lag 12 where it takes the value of 27.82%.

Kendall and Stuart (1976) shows that under the null hypothesis of serial independence the  $j$ -order autocorrelation has an expected value of  $-1/(T-j)$ . Because the Bartlett estimation is based on sample autocorrelations, it is obviously biased. The expected value of the Bartlett variance ratio is straightforward,

$$E[VR] = E\left[1 + 2\sum_{j=1}^{k-1} \frac{k-j}{k} \rho_j\right] = 1 + 2\sum_{j=1}^{k-1} \frac{k-j}{k} \left(-\frac{1}{T-j}\right) = \frac{2-k}{k} + \sum_{j=1}^{k-1} \frac{T-k}{T-j}. \quad (15)$$

Dividing Equation (13) by the expected value of the VR of a serially independent sequence, an unbiased estimation can be derived. Figure 7 shows that the Cochrane and the unbiased Bartlett estimations are almost identical; the correlation between the two measures is 99.83%.

Lo and MacKinlay (1988) show that the variance ratio is asymptotically normally distributed under the  $H_0$  of random walk of the form

$$VR_k \sim N\left(1, \sqrt{\frac{2(2k-1)(k-1)}{3Tk^2}}\right). \quad (16)$$

Using Lo and MacKinlay results we find that the variance ratios are significantly different from unity so the random walk hypothesis can be rejected at any usual significance level. The empirical findings of the stock market are different from our results. First of all, the role of the random walk component is more important in stock prices. Lo and MacKinlay (1988) finds that price formation based on the random walk hypothesis can be rejected applying the Center for Research in Securities Prices (CRSP) value weighted stock index from 1962 to 1985 and they

present positive autocorrelations even up to 16 weeks. Their results are more significant for small stocks.

The art returns are negatively autocorrelated both at shorter and longer periods as well. The negative autocorrelation is striking because there are many justifications which would rather induce positive autocorrelation. The infrequent trading which is a particular characteristic of the art market should result in positive autocorrelation because the information can be impounded slowly into the prices. Furthermore, Chen and Rhee (2009) present empirical evidence from the stock market that short sales contribute to market efficiency by increasing the speed of price adjustment so short sales constraints might cause inefficiency on the art market.

Lo and MacKinlay (1988) state that positive autocorrelation in US small stock returns can be related to infrequent trading. The survivorship bias also should result in positive autocorrelations since investors condition their sales on price appreciation. If the price of a work of art dropped significantly, its owner is likely to hold his item, for example, many impressionist paintings have dropped out of the market after they were sold to Japanese investors for record prices just before the burst of the Japanese stock market bubble in 1989. However, as Mei and Moses (2002) state the survivorship is mitigated by two facts: auction houses have to sell estates independently of market prices and the auctioneers have incentives to sell inexpensive pieces by established artists to attract first time collectors.

The 'irrational exuberance' also can cause negative autocorrelation in art returns (see Pesando and Shum, 2007); investors can bid too aggressively in certain auctions; however, these sudden run-ups in prices are likely to revert on the subsequent auctions of the works of art.

Figures 3-7 indicate that the art index has two components: a random walk process which accounts for about 28% of the variance of art returns and a stationary process which accounts for

the remaining variance and which has impact approximately up to lag 12. One possible reasoning for our results can be that art returns have two components: a common and an idiosyncratic component and the later is a white noise process.

We fit different autoregressive moving average process on art returns and select the ones which are not too far from their long-run value at lag 12 and their variance ratio is close to 28% at lag 12. The selection procedure is as follows:

- first, ARMA processes with variance ratio out of the one standard error band of the art price variance ratio are dropped out, then
- models with significant Ljung-Box Q statistics<sup>9</sup> are deselected and finally
- the difference between the variance ratio at lag 12 and the long run variance ratio is investigated and only those models are short listed where the difference is within one standard error of the art price variance ratio at lag 12.

The selected models and their diagnostic statistics can be found in Table 1.

[ Please insert Table 1 here ]

We find 36 candidate ARMA models for describing the art returns. The selected ARMA models are simulated by Monte Carlo experiment and the variance ratios are calculated by various methods. First, we use Cochrane's estimation described in Equation (13). Then, we expand our analysis with different kernels frequently used in the literature. Andrews' (1991) paper deals with the properties of different kernel functions for estimating spectral density. He estimates spectral density at frequency zero by Bartlett, Parzen, Tukey-Hanning and the quadratic spectral (QS) kernels. We estimate the variance ratios by the same kernels as in

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<sup>9</sup> Significant Q statistics can be the sign of misspecified ARMA modeling.

Andrews (1991) and weigh the  $k$ -order autocorrelations of the first difference of the real log art index of the form

$$\sigma_k^2 = \left( 1 + 2 \sum_{j=1}^{k-1} \kappa \rho_j \right) \sigma_{\Delta P}^2, \quad (17)$$

where  $\kappa$  is the kernel function and it is equal to

$$\text{Bartlett:} \quad \kappa_{BT} = \begin{cases} 1 - |x| \leq 1, \\ 0 \text{ otherwise} \end{cases}$$

$$\text{Parzen:} \quad \kappa_{PR} = \begin{cases} 1 - 6x^2 + 6|x^3| \text{ for } 0 \leq |x| \leq 1/2, \\ 2(1 - |x|)^3 \text{ for } 1/2 \leq |x| \leq 1, \\ 0 \text{ otherwise} \end{cases}$$

$$\text{Tukey Hanning:} \quad \kappa_{TH} = \begin{cases} (1 + \cos(\pi x)) / 2 \text{ for } |x| \leq 1, \\ 0 \text{ otherwise} \end{cases}$$

$$\text{Quadratic spectral:} \quad \kappa_{QS} = \frac{25}{12(\pi x)^2} \left( \frac{\sin(6\pi x / 5)}{6\pi x / 5} - \cos(6\pi x / 5) \right), \text{ where}$$

$x=j/k$ . Andrews finds that the quadratic spectral (QS) kernel is optimal in terms of Mean Squared Error (MSE) and the Bartlett kernel is inferior with respect to the others. On the other hand, Newey and West (1994) find no evidence preferring the QS or any other kernels and they argue that only bandwidth selection matters and the choice of kernel is of secondary importance.

After the selection procedure we simulate the candidate ARMA processes plus a white noise process which has the same variance as the art returns' and check whether or not the art price variance ratios up to lag 30 is within the one standard error band of the simulated variance ratios.

We find that in terms of the Mean Average Squared Error (MASE) the Cochrane method performs best.<sup>10</sup>

We choose the best model for describing art prices based on the  $R$  squares of regressions of art price variance ratios on simulated variance ratios. There are nine ARMA models out of the 36 simulated ones which can replicate the behavior of art price variance ratios. They are summarized in Table 2.

[ Please insert Table 2 here ]

The models in Table 2 can almost fully replicate the API variance ratios and they are statistically not different from the art price variance ratios. The best model is the ARMA(2,11) based on the  $R$  squares. The variance ratios of this model and the associated standard errors can be found in Figure 8.

[ Please insert Figure 8 here ]

The variance ratios of the ARMA(2,11) model declines up to  $k=12$  where it is 35%, in contrast with 28% which is the variance ratio of the API at the same lag; however, the difference is not significant.

Care should be taken when interpreting the rejection of the random walk hypothesis as the rejection of efficient pricing. Lucas (1978) shows that the rational asset prices do not need to follow martingale sequence, of which the random walk is a special case. Therefore, it is not necessary to interpret the rejection of random walk as market inefficiency, although the results may be interpreted as a rejection of some economic model of efficient price formation.

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<sup>10</sup> The average MASE is calculated of the 36 simulated ARMA and the white noise process. These results are available upon request.

One can imagine an art market which is not perfectly frictionless and the prices differ from fundamental values as results of shocks. If the shocks have persistent effect on prices, for example because they follow an autoregressive process, it implies negative autocorrelations. This assumption has a theoretical counterpart on the stock market. Summers (1986) shows that stock prices not necessarily reflect rational valuations and they can take long swings from fundamental values; however, asset prices return to the fundamental value in the long run which generates negative autocorrelation in the returns.

Why is the transitory component so important? Works of art are typical long run investments because of huge transaction costs (up to 20-30% of the market value considering the buyer's and the seller's premium) and low liquidity. Mean reversion in returns have crucial impact on portfolio decision since long run investments may have time diversification effects. Samuelson (1963) shows that independent repeated games induce increasing variance measured in dollars. If prices are mean reverting, that is, showing negative autocorrelation in the long run, it is rational to invest more with long investment horizons because of time-diversification effects. On the other hand, if there is significant transitory component in asset prices, rational economic agents are reluctant to tie decision to current market valuations. An investment strategy that suggests to buy when prices have dropped significantly and to sell when prices have surged significantly makes sense on a market where prices have a transitory component.

Another possible explanation can be the overreaction of collectors as Mei and Moses (2005) and Pesando and Shum (2007) point out. This property can be in relation with the stylistic risk described in Goetzmann (1993). If there is a shift in investors' taste, it can cause large negative autocorrelations. Supposing a positive shock for a particular work of art, the demand increases dramatically and investors may overreact; however, when collectors focus on another work of

art, valuation may drop significantly generating lower returns and negative autocorrelations in realized returns.

Another interpretation for the temporary component in art prices can be that experts who set the price estimates or investors are subjected to ‘rule of thumb learning’, as they anchor their current valuations to the previous sales price (see Beggs and Graddy, 2005).

## **5. Time varying components**

We have reasons to assume that price formation of art works is not stationary or at least the random walk component should be more important as it is studied on the stock market. We check three possible structural breakpoints, 1935, 1945 and 1973.

Before 1935 the data are sparse and this might have undesirable effect on our estimation. Mei and Moses (2002) argue that the lack of data points can induce spurious negative correlations in art returns. It is well-documented in the literature that asset prices follow different processes in the pre- and in the post-World War II eras. Fama and French (1988), for example, measure a more significant stationary component in stock prices for the period before 1940. And finally, we investigate the post 1973 period because there is a significant change in the institutional background of the art market, as price estimates and their wide publication are introduced in 1973. Since then the major auction houses have provided a low and a high estimate of the value of each lot. Mei and Moses (2005) argue that these estimates have economically and statistically important effect on art prices because auctioneers’ price estimates are biased upward and investors are credulous consistent with studies on stock markets. Higher high estimate results in higher sales price and generates lower future returns. They argue that upward bias is persistent over a period of 30 years because of agency problems and rational learning cannot eliminate it.

[ Please insert Figure 9 here ]

Figure 9 shows the variance ratios up to lag 30 of art prices for the period of 1875-1934. We measure somewhat lower random walk component for the data excluding years after 1934; however, the difference is within two standard errors so statistically not significant. The variance ratio declines up to lag 12 where it takes the value 22% which is six percentage points lower than the same measure for the longer times series. We can conclude that before 1935 art prices have the same characteristics as for the whole period. We have to point it out that the possible bias caused by sparse data before 1935 can drive the results for both sample periods.

Frey and Pommerehne (1989) argue consistent with our results that up to World War II, superior knowledge of art, in particular expert knowledge of paintings, could have helped to achieve a higher return than someone who picked auctioned paintings simply by chance. Systematic winners could be those ones who can influence collector's taste; however, there is no evidence of the existence of this capability.

[ Please insert Figure 10 here ]

Figure 10 shows that the art market has significantly higher random walk component after 1935. Two explanations can be imagined:

- the estimation before 1935 is not reliable because of sparse data in the Mei-Moses Fine Art Index but in this case the random walk component is biased downward for the full sample size.
- there is a structural change in the art market after World War II and it has a significant effect on the estimation for the period of 1935-2008.

As the estimation is quite noisy probably because of the small sample size we cannot find a lag from which the variance ratio can be thought constant. The variance ratio has a minimum

value of 61% at  $k=21$  which is almost two standard errors from the same measure for the whole sample.<sup>11</sup> A fairly good estimate for the weight of the random walk component is 70-75%. The associated standard errors are large; it is partly because of the small sample size and partly because of the large random walk component<sup>12</sup>. The magnitude of standard errors is too large for our results to be conclusive on random walk hypothesis; however, we can confirm that price formation is significantly different before and after 1935.

Figure 11 shows the variance ratios at different lags for the period 1875-1944; extending the data by ten years does not change the extent of the random walk component. It is not significantly different either from the measure for the period of 1875-1934 or from the measure of the whole sample.

[ Please insert Figure 11 here ]

We perform the Cochrane estimator also for the period from 1945 to 2008 and find that the size of the random walk component is even larger than the one for the period from 1935 to 2008 (see Figure 12). The past 64 years of the art prices are characterized by random walk process which means the weak-form efficiency of the art market cannot be rejected.

[ Please insert Figure 12 here ]

Finally, we check the last possible structural breakpoint: 1973. Figure 13 shows that the estimation excluding the post-1973 period almost coincides with the estimation for the whole sample.

[ Please insert Figure 13 here ]

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<sup>11</sup> We perform estimation by all the other estimators in Andrews (1991) and we find similar results which are available upon request. The Parzen and the Tukey estimator is less noisy but shows similar characteristics as the Cochrane method, the VR declines sharply till  $k=3$  and then they swing between 70-75%.

<sup>12</sup> The asymptotic standard errors grow in the function of size of the random walk component since  $\sigma(VR) = VR * (4k / 3T)^{0.5}$ .

The last 36 years of the art market can be described by random walk hypothesis (see Figure 14). The variance ratio does not decline below unity, so we can conclude that weak-form of market efficiency has been held since at least 1973.

[ Please insert Figure 14 here ]

It is hard to detect the year from which the market becomes efficient as at least since 1945 we cannot reject the random walk hypothesis. The size of the random walk component increases with the starting year of our analysis: the later we start, the larger the random walk component is. The smaller permanent component before 1935 can be due to poor data quality since if we include the years before 1935, no matter when we finish our analysis, the results are very similar. The introduction of price estimates in 1973 does not cause inefficiency in the art market; however, we know the estimates are biased. One can say that despite the biased estimates more investors have been attracted to the market. One of the aims of the launch of price estimates was just the attraction of small individual collectors. So we believe, although the price estimates are biased, helped to establish an efficient market with flooding the market with new participants. Besides this factor there are many reasons for why the market changes in the post-war era. Before World War II the art market was featured with some rich nobles and tycoons who collected works of art and some art dealers. In the post-war era along with the spread of open-economies and globalization, the market became easier to reach by small investors.

Care should be taken interpreting our results for the whole US art market; however, Candela and Scorcu (2001) investigate the dealers' market and find that the prices are non-stationary, they are integrated of order one, which is not contradictory to our findings, at least not from 1945. On the other hand, we have to note based on Candela and Scorcu (2001) that prices are more unstable on the auction market because collectors might be driven by euphoria causing

aggressive bidding and winners' course. The severe price differences between the two markets might be eliminated by arbitrage transactions by dealers on both markets. If we analyze the prices of both markets sequentially, there is a casual link from the auction market to the dealers' market as auction prices represent a benchmark for galleries (see Candela and Scorcu, 2001). As another point of view, Hutter et al. (2007) find that the aggregate price level of the dealers' market is higher than that of the auction market in contemporary arts. The yearly price developments show similar upward patterns for both dealer and auction markets, but dealer prices are stickier in their downward movements.

## **6. Concluding remarks**

We find that in our sample period of 1875-2008 the US art prices follow neither a pure random walk, nor a pure stationary process. The Cochrane (1988) variance ratio declines up to lag 12 where it takes the value of 29.43% which is more than two standard errors from unity and zero so we can reject either a null hypothesis of pure random walk or a null of pure stationary process. Based on the studied ARMA processes, art prices are closest to an ARMA(2,11).

We identify structural breakpoints in our data testing 1935, 1945 and 1973. Before 1935 the RSR estimation of Mei Moses Fine Art Index is based only on a few data points which can result in spurious negative autocorrelation (see Mei and Moses, 2002). The choice of 1945 as the starting date of the post-war era is an obviously possible structural breakpoint, which is commonly used in the literature. Finally, we choose 1973 because of the introduction of price estimates. Mei and Moses (2005) argue that these estimates are upward biased; therefore, they potentially influence the art price formation. If we exclude the first 60 years from our data, the random walk hypothesis cannot be rejected. The first 60 years of the dataset; however, exhibit

similar result to the one obtained for the whole sample period. There is no significant difference between our results, if we choose 1935, 1945 or 1973 as the starting point of our estimates. If we choose 1935 as the starting date, the random walk component is approximately 75%, on which we cannot reject the random walk hypothesis. If we start our test with 1945 or 1973, we obtain an even higher random walk component; however, the standard errors are much higher, partly because of the higher random walk component, and partly because of the smaller sample size. These estimations are statistically not different, and based on none of them can we reject the null hypothesis of random walk.

The significant difference of art price formation before and after 1935 have many possible causes. One is that the art market has experienced institutional changes after 1935. Along with the globalization and the introduction of price estimates the market was expanding, new investors appeared, so the structural change is an established explanation. The other one can be the unreliable price estimates in the RSR regressions before 1935 which cause spurious negative autocorrelation in art returns and therefore smaller variance ratios.

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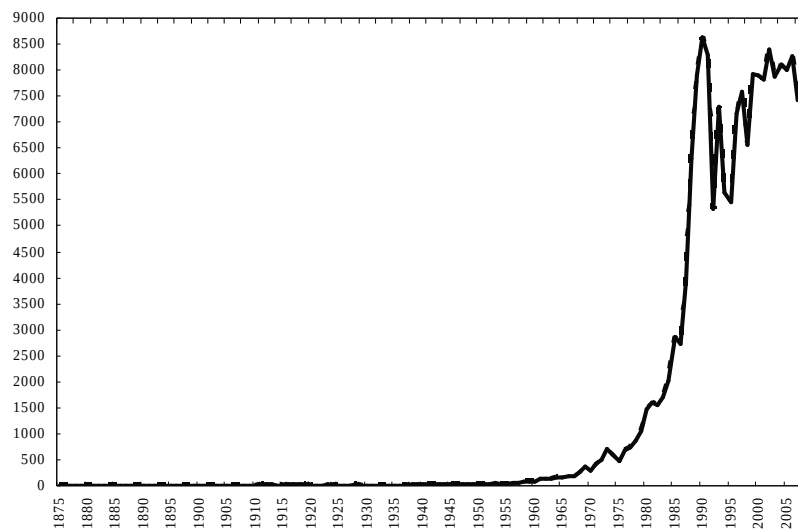
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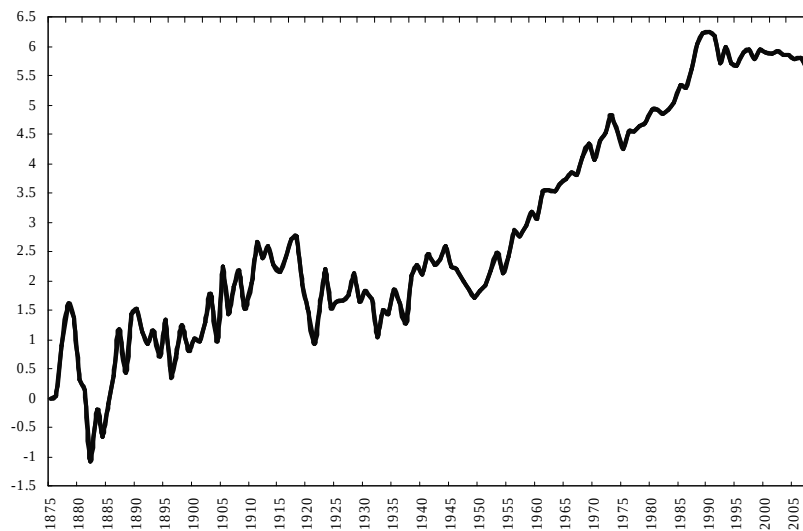
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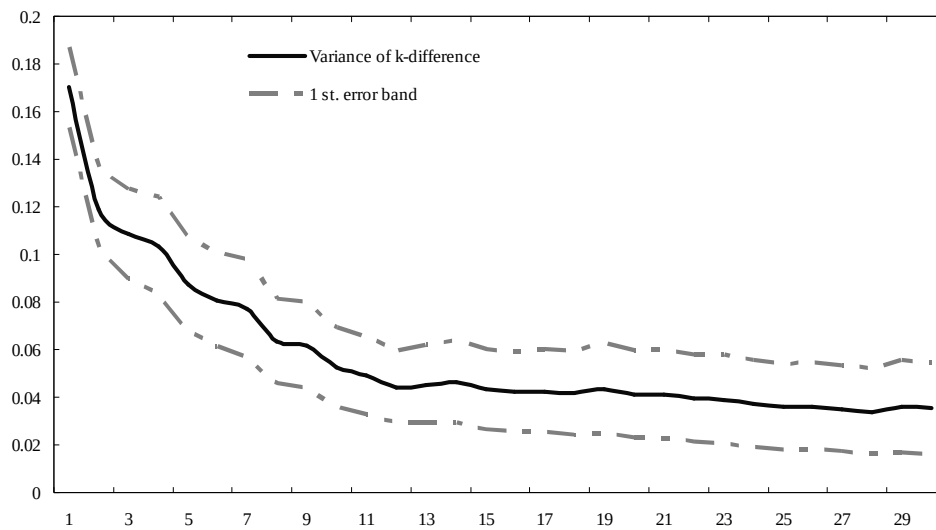
## Figures



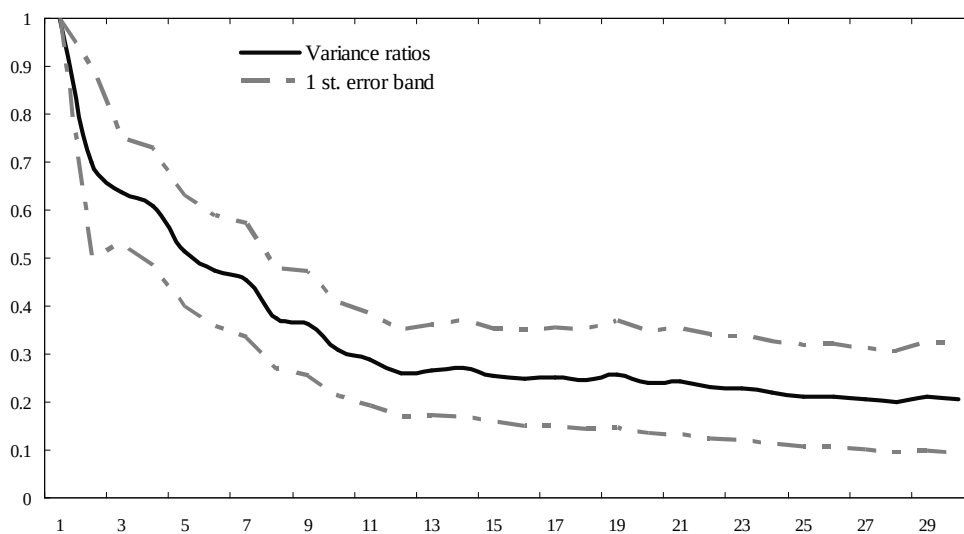
**Figure 1** The figure shows the historical price index of the US art market from 1875 to 2008



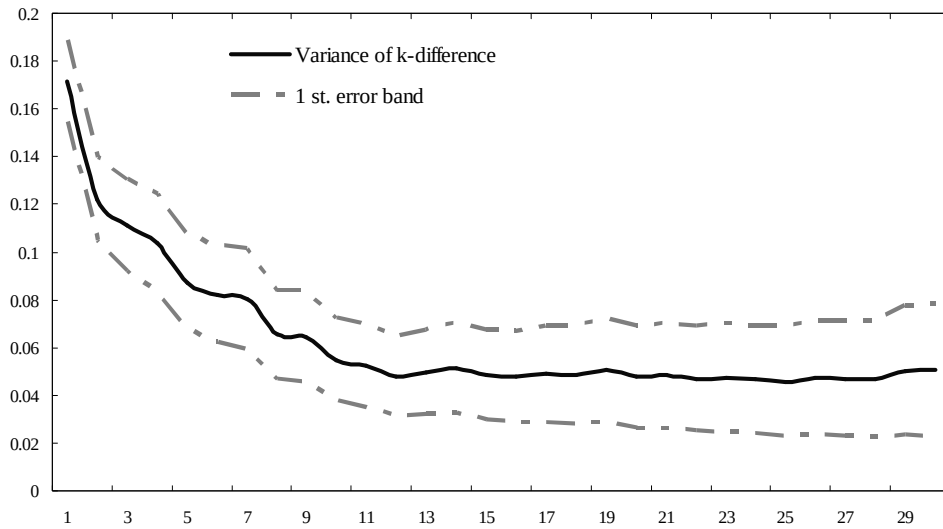
**Figure 2** The figure shows the logarithm of historical real price index of the US art market from 1875 to 2008



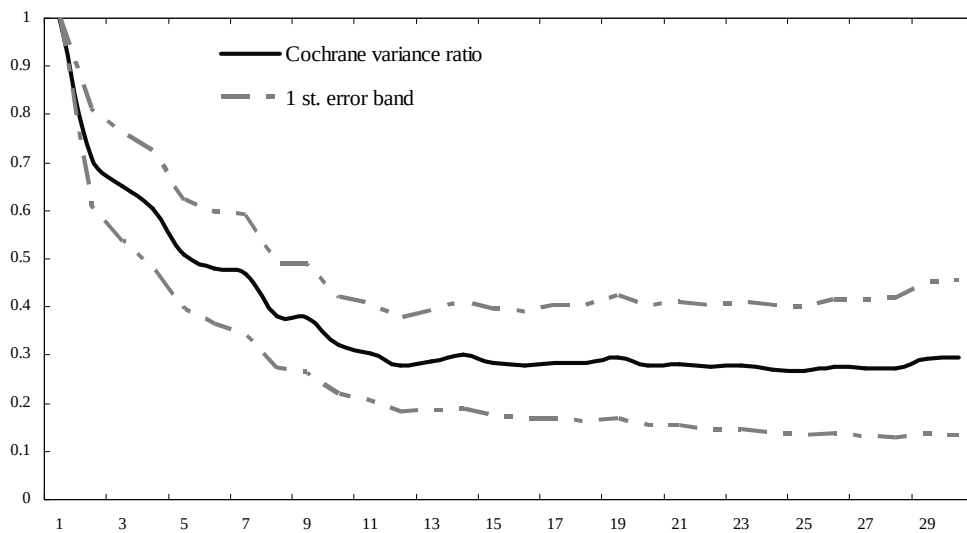
**Figure 3** The figure shows the variance of  $k$ -year returns of the art index and the asymptotic standard errors. The estimation is conducted by Bartlett kernel (see e.g., Cochrane, 1988). The asymptotic standard errors are calculated by  $\sigma(\sigma_k^2) = \sigma_k^2 * (4k/3T)^{0.5}$ .



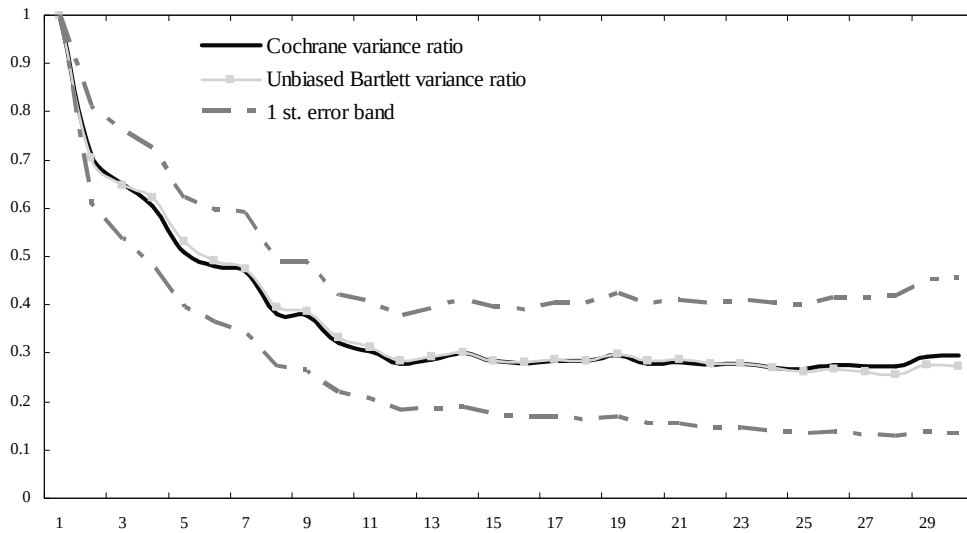
**Figure 4** The figure shows the variance ratios of  $k$ -year returns of the art index and the asymptotic standard errors. The estimation is conducted by Bartlett kernel (see e.g., Cochrane, 1988). The asymptotic standard errors are calculated by  $\sigma(VR) = VR * (4k/3T)^{0.5}$ .



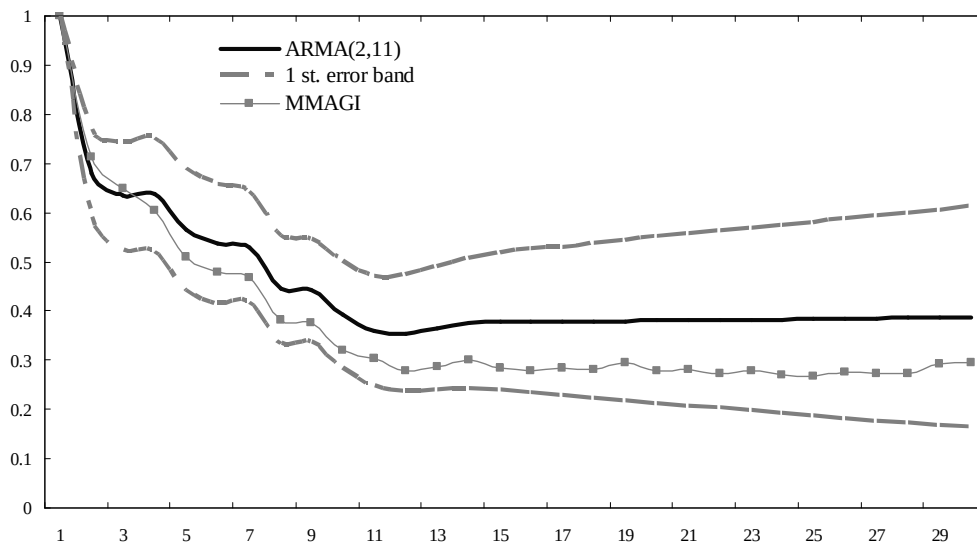
**Figure 5** The figure shows the variances of  $k$ -year returns of the art index and the asymptotic standard errors. The estimation is conducted by the Cochrane method. The asymptotic standard errors are calculated by  $\sigma(\sigma_k^2) = \sigma_k^2 * (4k / 3T)^{0.5}$ .



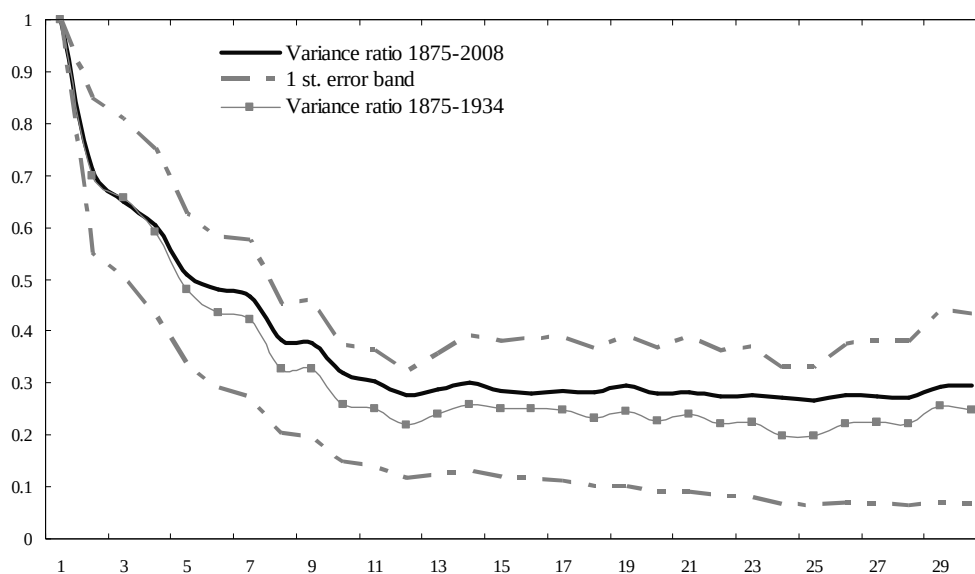
**Figure 6** The figure shows the variance ratios (VR) of  $k$ -year returns of the art index and the asymptotic standard errors. The estimation is conducted by the Cochrane method. The asymptotic standard errors are calculated by  $\sigma(\text{VR}) = \text{VR} * (4k / 3T)^{0.5}$ .



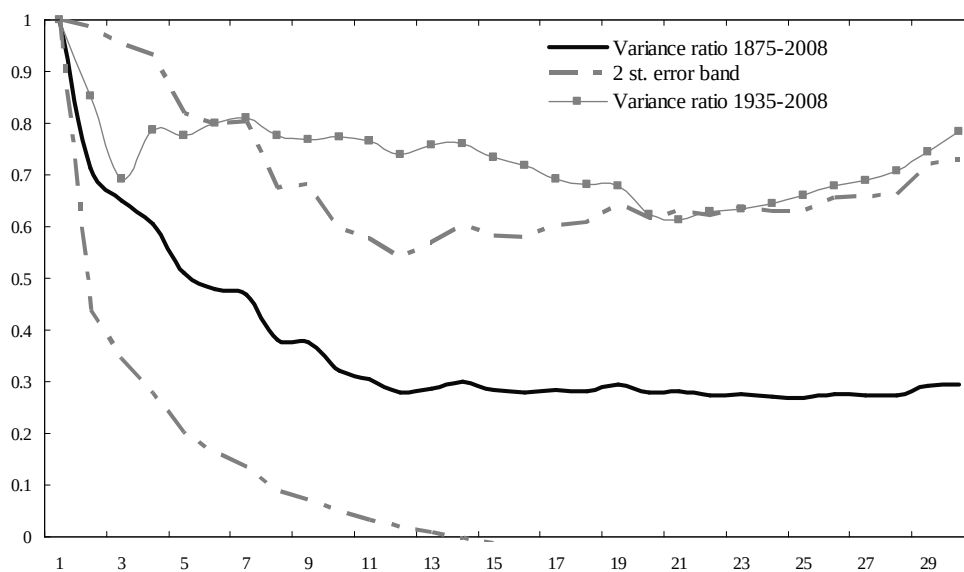
**Figure 7** The Cochrane and the adjusted Bartlett variance ratios (VR) and the asymptotic standard errors of the Cochrane ratios. The asymptotic standard errors are calculated by  $\sigma(\text{VR}) = \text{VR} * (4k / 3T)^{0.5}$ .



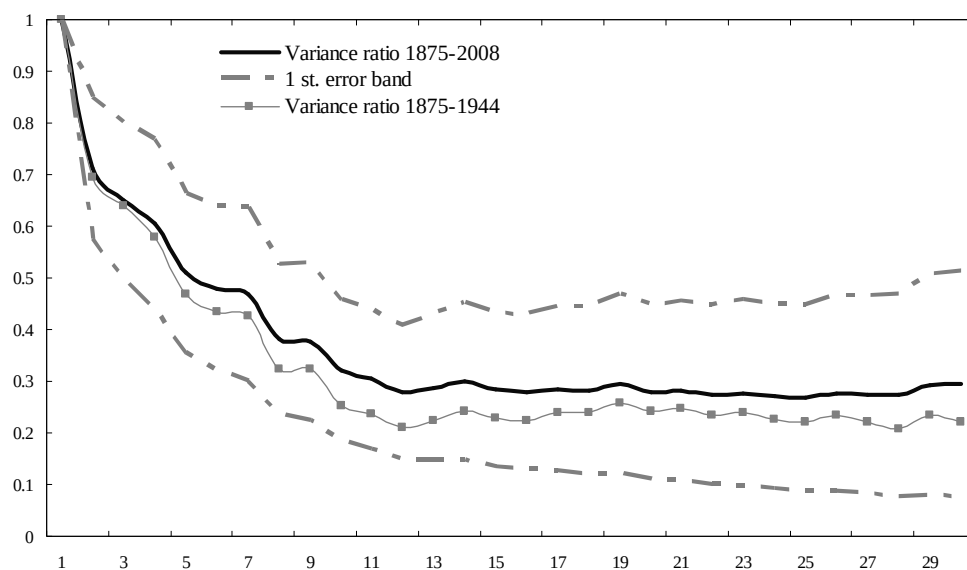
**Figure 8** shows the variance ratios and associated Monte Carlo one-standard-error-band of the simulated ARMA(2,11) model. The variance ratios of the Art Price Index are also drawn in the figure. The estimations are performed by Cochrane's (1988) procedure.



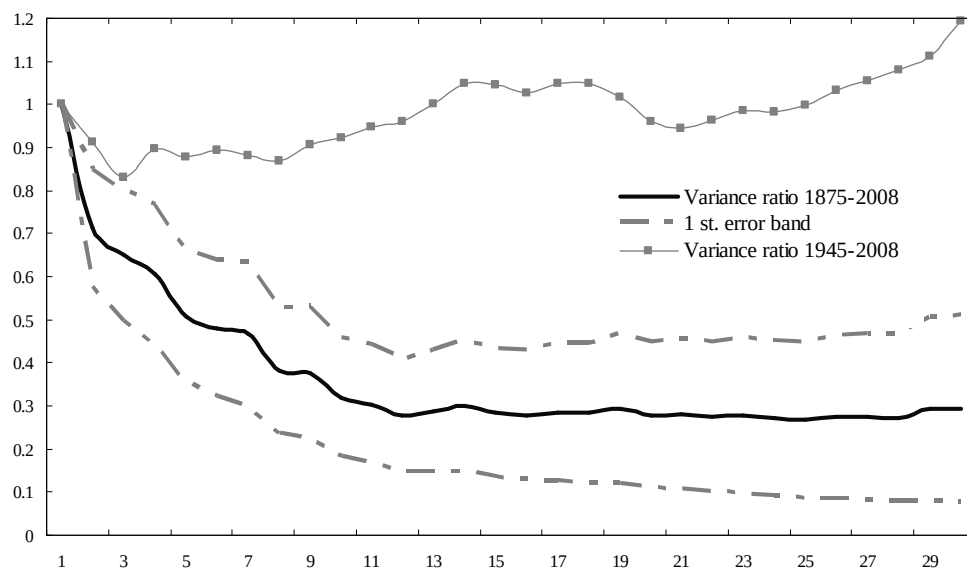
**Figure 9** shows the variance ratios (VR) of the Art Price Index for the period 1875-2008 and 1875-1934. The asymptotic one standard error band belongs to the VRs for the period 1875-1934. The estimations are performed by Cochrane's (1988) procedure. The asymptotic standard errors are calculated by  $\sigma(VR) = VR * (4k / 3T)^{0.5}$ .



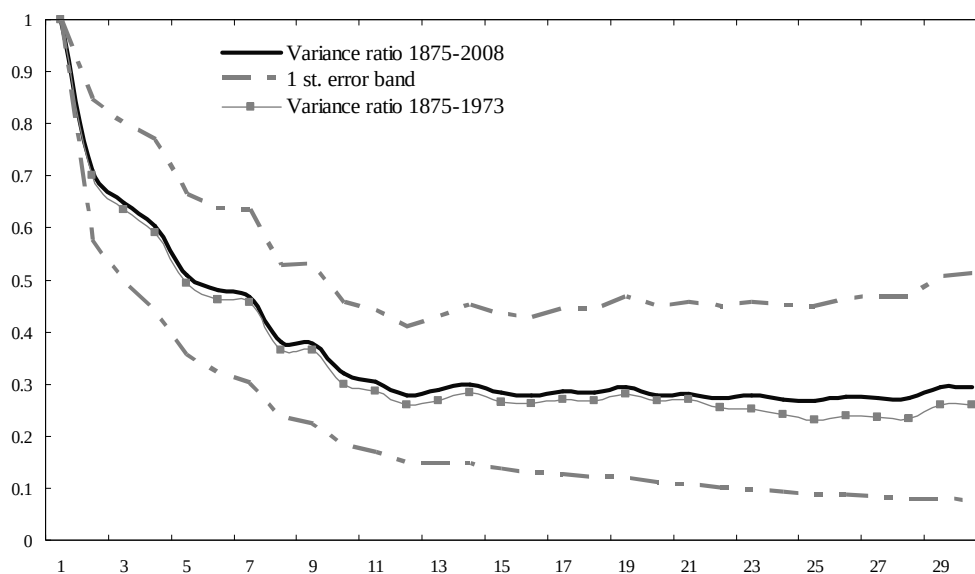
**Figure 10** shows the variance ratios of the Art Price Index for the period 1875-2008 and 1935-2008. The asymptotic two standard error band belongs to the variance ratios for the period 1875-2008. The estimations are performed by Cochrane's (1988) procedure. The asymptotic standard errors are calculated by  $\sigma(VR) = VR * (4k / 3T)^{0.5}$ .



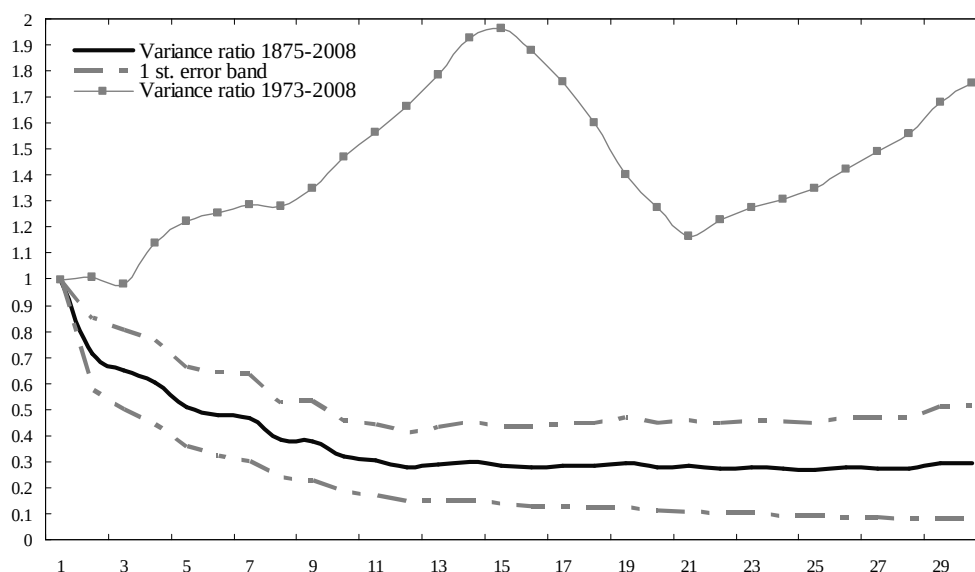
**Figure 11** shows the variance ratios of the Art Price Index for the period 1875-2008 and 1875-1944. The asymptotic one standard error band belongs to the variance ratios for the period 1875-2008. The estimations are performed by Cochrane's (1988) procedure. The asymptotic standard errors are calculated by  $\sigma(VR) = VR * (4k / 3T)^{0.5}$ .



**Figure 12** shows the variance ratios of the Art Price Index for the period 1875-2008 and 1945-2008. The asymptotic one standard error band belongs to the variance ratios for the period 1875-2008. The estimations are performed by Cochrane's (1988) procedure. The asymptotic standard errors are calculated by  $\sigma(VR) = VR * (4k / 3T)^{0.5}$ .



**Figure 13** This figure shows the variance ratios of the Art Price Index for the period 1875-2008 and 1875-1973. The asymptotic one standard error band belongs to the variance ratios for the period 1875-2008. The estimations are performed by Cochrane's (1988) procedure. The asymptotic standard errors are calculated by  $\sigma(VR) = VR * (4k / 3T)^{0.5}$ .



**Figure 14** This figure shows the variance ratios of the Art Price Index for the period 1875-2008 and 1973-2008. The asymptotic one standard error band belongs to the variance ratios for the period 1875-2008. The estimations are performed by Cochrane's (1988) procedure. The asymptotic standard errors are calculated by  $\sigma(VR) = VR * (4k / 3T)^{0.5}$ .

## Tables

| ARMA(i,j)      | (1,9) | (1,11) | (2,7) | (2,10) | (2,11) | (2,15) | (3,10) | (3,11) | (4,7)  | (4,18) | (5,10) | (5,12) | (6,7)  | (6,8)  | (6,9) | (7,1)  | (7,3)  | (7,4)  |
|----------------|-------|--------|-------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|-------|--------|--------|--------|
| R <sup>2</sup> | 0.38  | 0.38   | 0.27  | 0.36   | 0.35   | 0.46   | 0.36   | 0.37   | 0.33   | 0.46   | 0.41   | 0.37   | 0.29   | 0.37   | 0.38  | 0.30   | 0.34   | 0.35   |
| SIC            | 0.90  | 0.96   | 1.01  | 0.97   | 1.01   | 0.95   | 0.99   | 1.00   | 0.98   | 1.07   | 0.91   | 1.04   | 1.05   | 0.97   | 0.97  | 0.81   | 0.82   | 0.83   |
| Q              | 30.50 | 25.34  | 31.51 | 21.33  | 21.31  | 20.89  | 23.61  | 23.31  | 27.04  | 19.58  | 29.84  | 24.44  | 29.77  | 29.06  | 27.90 | 35.78  | 36.83  | 33.51  |
| p(Q)           | 0.06  | 0.12   | 0.09  | 0.32   | 0.26   | 0.10   | 0.21   | 0.18   | 0.21   | 0.05   | 0.05   | 0.11   | 0.12   | 0.11   | 0.11  | 0.15   | 0.08   | 0.12   |
| VR(12)         | 0.33  | 0.35   | 0.33  | 0.50   | 0.50   | 0.31   | 0.42   | 0.48   | 0.42   | 0.42   | 0.44   | 0.59   | 0.26   | 0.41   | 0.34  | 0.41   | 0.30   | 0.33   |
| VR(30)         | 0.32  | 0.46   | 0.29  | 0.41   | 0.40   | 0.38   | 0.39   | 0.39   | 0.44   | 0.42   | 0.46   | 0.43   | 0.32   | 0.47   | 0.43  | 0.46   | 0.34   | 0.38   |
| VR( $\infty$ ) | 0.32  | 0.46   | 0.29  | 0.41   | 0.40   | 0.39   | 0.30   | 0.39   | 0.42   | 0.45   | 0.47   | 0.44   | 0.32   | 0.47   | 0.43  | 0.47   | 0.34   | 0.39   |
| ARMA(i,j)      | (8,2) | (8,3)  | (9,1) | (9,3)  | (9,8)  | (10,3) | (10,6) | (11,5) | (11,6) | (12,2) | (12,4) | (12,7) | (12,8) | (19,3) | (0,9) | (0,10) | (0,17) | (0,19) |
| R <sup>2</sup> | 0.30  | 0.34   | 0.27  | 0.30   | 0.36   | 0.33   | 0.36   | 0.38   | 0.37   | 0.34   | 0.39   | 0.43   | 0.42   | 0.33   | 0.24  | 0.23   | 0.29   | 0.38   |
| SIC            | 0.84  | 0.81   | 0.89  | 0.91   | 0.96   | 0.89   | 0.93   | 0.90   | 0.95   | 0.88   | 0.86   | 0.88   | 0.92   | 1.10   | 1.06  | 1.10   | 1.22   | 1.14   |
| Q              | 35.17 | 35.68  | 40.06 | 33.57  | 31.99  | 37.32  | 33.47  | 32.14  | 32.31  | 25.29  | 22.10  | 22.16  | 21.98  | 16.54  | 30.46 | 30.81  | 19.91  | 19.21  |
| p(Q)           | 0.13  | 0.10   | 0.07  | 0.15   | 0.06   | 0.07   | 0.07   | 0.12   | 0.09   | 0.56   | 0.63   | 0.45   | 0.40   | 0.92   | 0.08  | 0.06   | 0.10   | 0.06   |
| VR(12)         | 0.28  | 0.40   | 0.32  | 0.39   | 0.34   | 0.39   | 0.35   | 0.37   | 0.36   | 0.57   | 0.50   | 0.52   | 0.54   | 0.47   | 0.31  | 0.31   | 0.26   | 0.28   |
| VR(30)         | 0.32  | 0.47   | 0.38  | 0.41   | 0.44   | 0.45   | 0.44   | 0.44   | 0.44   | 0.46   | 0.37   | 0.46   | 0.47   | 0.48   | 0.31  | 0.31   | 0.45   | 0.47   |
| VR( $\infty$ ) | 0.31  | 0.48   | 0.38  | 0.39   | 0.44   | 0.46   | 0.44   | 0.45   | 0.45   | 0.46   | 0.43   | 0.47   | 0.48   | 0.46   | 0.31  | 0.31   | 0.45   | 0.47   |

**Table 1** shows the  $R^2$ s, the Schwarz information criteria (SIC), the Ljung-Box  $Q$  statistics and their  $p$ -values and the theoretical values of the variance ratios (VR) at lag 12, 30 and at infinity of the ARMA models of art returns. The theoretical values of variance ratios are calculated by  $VR_k = \left( \sum_{j=0}^k a_j \right)^2 / \sum_{j=0}^k a_j^2$ , where  $a_j$ s are the coefficients of the moving average representation of each ARMA model. The  $R^2$ s are adjusted with degrees of freedom.

| kARMA(i,j) | MMAGI |       | (0,9) |       | (0,10) |       | (0,19) |       | (1,11) |       | (2,10) |       | (2,11) |       | (2,15) |       | (3,10) |       | (3,11) |       |
|------------|-------|-------|-------|-------|--------|-------|--------|-------|--------|-------|--------|-------|--------|-------|--------|-------|--------|-------|--------|-------|
|            | VR    | stdev | VR    | stdev | VR     | stdev | VR     | stdev | VR     | stdev | VR     | stdev | VR     | stdev | VR     | stdev | VR     | stdev | VR     | stdev |
| 1          | 1.00  | 0.00  | 1.00  | 0.00  | 1.00   | 0.00  | 1.00   | 0.00  | 1.00   | 0.00  | 1.00   | 0.00  | 1.00   | 0.00  | 1.00   | 0.00  | 1.00   | 0.00  | 1.00   | 0.00  |
| 2          | 0.71  | 0.10  | 0.76  | 0.10  | 0.75   | 0.10  | 0.74   | 0.10  | 0.75   | 0.10  | 0.68   | 0.09  | 0.68   | 0.09  | 0.67   | 0.14  | 0.64   | 0.08  | 0.65   | 0.09  |
| 3          | 0.65  | 0.11  | 0.74  | 0.12  | 0.73   | 0.12  | 0.66   | 0.12  | 0.69   | 0.13  | 0.64   | 0.12  | 0.63   | 0.11  | 0.60   | 0.19  | 0.57   | 0.10  | 0.60   | 0.11  |
| 4          | 0.60  | 0.12  | 0.71  | 0.11  | 0.71   | 0.11  | 0.68   | 0.13  | 0.72   | 0.14  | 0.65   | 0.12  | 0.64   | 0.12  | 0.62   | 0.18  | 0.61   | 0.10  | 0.63   | 0.12  |
| 5          | 0.51  | 0.11  | 0.57  | 0.11  | 0.57   | 0.11  | 0.60   | 0.15  | 0.65   | 0.15  | 0.58   | 0.14  | 0.57   | 0.12  | 0.51   | 0.15  | 0.52   | 0.12  | 0.55   | 0.13  |
| 6          | 0.48  | 0.12  | 0.49  | 0.10  | 0.50   | 0.10  | 0.59   | 0.15  | 0.62   | 0.16  | 0.55   | 0.13  | 0.54   | 0.12  | 0.46   | 0.12  | 0.53   | 0.13  | 0.54   | 0.12  |
| 7          | 0.47  | 0.12  | 0.47  | 0.10  | 0.47   | 0.10  | 0.58   | 0.14  | 0.62   | 0.15  | 0.54   | 0.13  | 0.53   | 0.11  | 0.46   | 0.11  | 0.55   | 0.13  | 0.52   | 0.12  |
| 8          | 0.38  | 0.11  | 0.41  | 0.10  | 0.41   | 0.11  | 0.51   | 0.14  | 0.54   | 0.14  | 0.45   | 0.13  | 0.44   | 0.11  | 0.39   | 0.12  | 0.49   | 0.14  | 0.43   | 0.11  |
| 9          | 0.38  | 0.11  | 0.44  | 0.11  | 0.45   | 0.11  | 0.50   | 0.13  | 0.52   | 0.13  | 0.45   | 0.12  | 0.44   | 0.11  | 0.42   | 0.12  | 0.51   | 0.13  | 0.43   | 0.11  |
| 10         | 0.32  | 0.10  | 0.43  | 0.11  | 0.44   | 0.11  | 0.42   | 0.12  | 0.46   | 0.13  | 0.40   | 0.12  | 0.39   | 0.11  | 0.41   | 0.12  | 0.46   | 0.14  | 0.38   | 0.11  |
| 11         | 0.30  | 0.10  | 0.42  | 0.12  | 0.43   | 0.12  | 0.39   | 0.12  | 0.40   | 0.13  | 0.36   | 0.13  | 0.36   | 0.11  | 0.39   | 0.13  | 0.43   | 0.14  | 0.35   | 0.11  |
| 12         | 0.28  | 0.10  | 0.42  | 0.12  | 0.42   | 0.12  | 0.35   | 0.11  | 0.38   | 0.13  | 0.36   | 0.13  | 0.35   | 0.12  | 0.40   | 0.13  | 0.42   | 0.14  | 0.35   | 0.12  |
| 13         | 0.29  | 0.10  | 0.41  | 0.13  | 0.41   | 0.13  | 0.34   | 0.12  | 0.37   | 0.13  | 0.37   | 0.14  | 0.37   | 0.12  | 0.39   | 0.13  | 0.42   | 0.15  | 0.36   | 0.12  |
| 14         | 0.30  | 0.11  | 0.41  | 0.14  | 0.41   | 0.13  | 0.35   | 0.12  | 0.38   | 0.14  | 0.38   | 0.15  | 0.38   | 0.13  | 0.38   | 0.14  | 0.42   | 0.15  | 0.37   | 0.13  |
| 15         | 0.28  | 0.11  | 0.40  | 0.14  | 0.40   | 0.14  | 0.34   | 0.13  | 0.38   | 0.15  | 0.39   | 0.16  | 0.38   | 0.14  | 0.37   | 0.14  | 0.42   | 0.16  | 0.37   | 0.14  |
| 16         | 0.28  | 0.11  | 0.40  | 0.15  | 0.40   | 0.14  | 0.32   | 0.13  | 0.39   | 0.16  | 0.39   | 0.17  | 0.38   | 0.15  | 0.36   | 0.15  | 0.41   | 0.16  | 0.37   | 0.15  |
| 17         | 0.28  | 0.12  | 0.40  | 0.15  | 0.40   | 0.15  | 0.33   | 0.14  | 0.39   | 0.16  | 0.39   | 0.18  | 0.38   | 0.15  | 0.36   | 0.16  | 0.40   | 0.16  | 0.38   | 0.15  |
| 18         | 0.28  | 0.12  | 0.39  | 0.16  | 0.40   | 0.15  | 0.33   | 0.15  | 0.39   | 0.17  | 0.39   | 0.18  | 0.38   | 0.16  | 0.36   | 0.16  | 0.40   | 0.17  | 0.38   | 0.16  |
| 19         | 0.29  | 0.13  | 0.39  | 0.16  | 0.39   | 0.16  | 0.34   | 0.15  | 0.40   | 0.18  | 0.39   | 0.19  | 0.38   | 0.16  | 0.37   | 0.17  | 0.39   | 0.17  | 0.38   | 0.17  |
| 20         | 0.28  | 0.12  | 0.39  | 0.17  | 0.39   | 0.16  | 0.35   | 0.16  | 0.40   | 0.19  | 0.40   | 0.20  | 0.38   | 0.17  | 0.37   | 0.18  | 0.39   | 0.18  | 0.38   | 0.17  |
| 21         | 0.28  | 0.13  | 0.39  | 0.17  | 0.39   | 0.16  | 0.36   | 0.17  | 0.40   | 0.19  | 0.40   | 0.21  | 0.38   | 0.17  | 0.38   | 0.18  | 0.39   | 0.18  | 0.38   | 0.18  |
| 22         | 0.27  | 0.13  | 0.39  | 0.18  | 0.39   | 0.17  | 0.36   | 0.18  | 0.41   | 0.20  | 0.40   | 0.21  | 0.38   | 0.18  | 0.38   | 0.19  | 0.38   | 0.18  | 0.38   | 0.19  |
| 23         | 0.28  | 0.13  | 0.39  | 0.18  | 0.39   | 0.17  | 0.37   | 0.18  | 0.41   | 0.21  | 0.40   | 0.22  | 0.38   | 0.19  | 0.38   | 0.19  | 0.38   | 0.19  | 0.38   | 0.19  |
| 24         | 0.27  | 0.13  | 0.38  | 0.19  | 0.39   | 0.18  | 0.38   | 0.19  | 0.41   | 0.21  | 0.40   | 0.22  | 0.38   | 0.19  | 0.37   | 0.20  | 0.38   | 0.19  | 0.39   | 0.20  |
| 25         | 0.27  | 0.13  | 0.38  | 0.19  | 0.38   | 0.18  | 0.38   | 0.20  | 0.41   | 0.22  | 0.40   | 0.23  | 0.38   | 0.20  | 0.37   | 0.20  | 0.38   | 0.20  | 0.39   | 0.20  |
| 26         | 0.28  | 0.14  | 0.38  | 0.20  | 0.38   | 0.18  | 0.39   | 0.21  | 0.42   | 0.23  | 0.41   | 0.24  | 0.38   | 0.20  | 0.37   | 0.21  | 0.38   | 0.20  | 0.39   | 0.21  |
| 27         | 0.27  | 0.14  | 0.38  | 0.20  | 0.38   | 0.19  | 0.39   | 0.22  | 0.42   | 0.24  | 0.41   | 0.25  | 0.38   | 0.21  | 0.37   | 0.21  | 0.38   | 0.20  | 0.39   | 0.22  |
| 28         | 0.27  | 0.14  | 0.38  | 0.21  | 0.38   | 0.19  | 0.39   | 0.22  | 0.42   | 0.24  | 0.41   | 0.25  | 0.39   | 0.21  | 0.38   | 0.22  | 0.38   | 0.21  | 0.39   | 0.22  |
| 29         | 0.29  | 0.16  | 0.38  | 0.21  | 0.38   | 0.19  | 0.40   | 0.23  | 0.42   | 0.25  | 0.41   | 0.26  | 0.39   | 0.22  | 0.38   | 0.22  | 0.37   | 0.21  | 0.39   | 0.23  |
| 30         | 0.29  | 0.16  | 0.38  | 0.22  | 0.38   | 0.20  | 0.40   | 0.24  | 0.42   | 0.25  | 0.41   | 0.27  | 0.39   | 0.22  | 0.38   | 0.23  | 0.37   | 0.21  | 0.39   | 0.23  |
| Rsqr       | -     | -     | 0.96  | -     | 0.97   | -     | 0.96   | -     | 0.96   | -     | 0.97   | -     | 0.98   | -     | 0.95   | -     | 0.93   | -     | 0.97   | -     |

**Table 2** shows the variance ratios (VR) and their standard deviation (stdev) of Art Price Index (API) and the nine simulated models which can replicate the behavior of the art index. In the last row the  $R$  squares can be found from the regressions of the API variance ratios on the variance ratios of the simulated models. The estimations are conducted by Cochrane's (1988) method.